

A New Minimalist Solution to the Seven Bridges of Königsberg

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Abstract

We record a one-line lemma: an Eulerian graph contains no bridges. As an immediate corollary, we obtain a new solution to the Seven Bridges of Königsberg problem.

Introduction

We work with finite connected multigraphs. For completeness we recall two standard definitions.

Definition 1 (Eulerian graph). *A connected graph is Eulerian if it admits a closed trail that traverses every edge exactly once.*

Definition 2 (Bridge). *An edge e of a connected graph G is a bridge if $G - e$ is disconnected.*

The Lemma

Lemma 1. *An Eulerian graph has no bridges.*

Proof. Let C be an Eulerian circuit in G . Every edge of G lies on C , hence on a cycle. No edge on a cycle can be a bridge. $\square$

Königsberg Bridges

Corollary 1. *There is no walk that traverses each of the seven bridges of Königsberg exactly once and returns to its starting point.*

Proof. Immediate from Lemma 1. $\square$

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References

- [1] Euler, Leonhard (1741). “Solutio problematis ad geometriam situs pertinentis”. *Commentarii Academiae Scientiarum Petropolitanae*: 128–140.